# Dr Georgina Maja Falster

## Personal information


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*Websites* Institutional profile; Google Scholar


## Career summary

I seek to understand how Earth's climate is changing in the context of natural variability, with a particular interest in the water cycle. I take a holistic approach to my research, combining information from modern observations, climate model simulations, and natural archives of long-term climate (palaeoclimate proxies). I use laboratory techniques to produce new water isotope and palaeoclimate proxy data. I use high-performance computing to run fully-coupled earth system models and analyse climate data using advanced statistical methods. I have experience with complex data visualisation, machine learning, and forward and inverse modelling methods in R and Python environments.

Teaching and communication are integral parts of my research ethos. I enjoy sharing my research through undergraduate teaching, postgraduate teaching, and researcher supervision, as well as scientific meetings and public outreach. I am dedicated to the academic and personal welfare of my students, and as a result have a proven record of teaching excellence.

## Professional appointments

2025– **Australian Research Council (ARC) DECRA Fellow**, *School of Physics, Chemistry and Earth Sciences, Adelaide University, Australia.* Producing and analysing new precipitation stable isotope, tree ring isotope, and climate model data for a holistic understanding of the Australian water cycle and how it is changing.

2021–2025 **ARC Centre of Excellence for Climate Extremes Research Fellow**, *Research School of Earth Sciences, Australian National University, Australia.* Investigating the drivers of Australian droughts.

2019–2021 **Postdoctoral Research Associate**, *Department of Earth and Planetary Sciences, Washington University in St. Louis, USA.* Using water isotope observations, proxy records, and model simulations to investigate forced and internal variability in ocean-atmosphere circulation.

2018–2019 **Marine Geoscientist**, *Geoscience Australia, Australia*

## Education

2015–2019 **PhD in Earth Science**, *The University of Adelaide, Australia* "Reconstructing Australia's Late Quaternary Climates from the Geochemistry of Lake Sediments and Snail Shells".
***Awarded Dean's Commendation for Doctoral Thesis Excellence.***

2008–2011 **Bachelor of Science with First Class Honours**, *The University of Adelaide, Australia*
***Awarded James Barrans Prize for top graduating earth science major.***

## Professional distinctions

### Major fellowships

2025–2028 Australian Research Council Discovery Early Career Researcher Award (DECRA) DE250100071 "*Quantifying Australia's long-term risk of rainfall extremes*". *AU$438,335*
***Sole palaeoclimate DECRA awarded in 2025.***

2025 L'Oréal-UNESCO For Women In Science Fellowship. *AU$25,000.*
***One of four awardees selected from an applicant pool >400 STEM researchers, and only the 6th earth scientist awarded a FWIS Fellowship since the award began in 2007.***

## Other competitive research funding (lead or sole CI)

2025–2026 National Computational Infrastructure (NCI) Merit Allocation Scheme "*The role of Indo-Pacific SST variability in global drought occurrence*". *1750 kSU on* Gadi *supercomputer, equivalent to AU$70,000 at official NCI rate.*

2026 Adelaide University Global Partnerships Engagement Fund & Past Global Changes Workshop grants to fund workshop "*Hydroclimate2k: a new database to understand water cycle variability through the Common Era"* , Canada, November 2026. *AU$32,000.*

2025 Adelaide University Environment Institute Research Funding & Research Equipment Support Schemes "*A state-of-the-art tree ring analytical facility to constrain decadal-scale risk in Australia's rainfall extremes*". *AU$40,130.*

2024 Past Global Changes (PAGES) grant to fund workshop "*Global-scale hydroclimate synthesis of the Common Era"*, Nottingham, 2024. *US$14,500.*

2023 Australian Institute of Nuclear Science & Engineering (AINSE) Early Career Researcher Award ALNGRA2205 "*Understanding the nature and drivers of spatial and temporal variability in the isotopic composition of precipitation across Australia*". *AU$10,000 + analyses in kind.*

2015–2019 AINSE Postgraduate Research Award ALNSTU11873 "*Assessing terrestrial climate variability over the last glacial-interglacial transition* ". *AU$30,000 + analyses in kind.*

## Selected academic awards & scholarships

2025 South Australian Tall Poppy award

2024 KSC Campbell Prize for Excellence in Earth Sciences teaching at the Australian National University

2023 ARC Centre of Excellence for Climate Extremes prize for best climate research published by an ECR

2019 Dean's Commendation for Doctoral Thesis Excellence
Fame-Lab Australia national semi-finalist, *Science communication competition*

2017 Esther Bright Prize, *Merit-based award to fund international conference attendance*; Australian Institute of Nuclear Science & Engineering (AINSE) International Travel Scholarship; University of Adelaide Research Abroad Scholarship

2016 Geological Society of Australia prize for best student presentation (oral) *at the Australian Earth Sciences Convention*
Adelaide University Graduate Union R. C. Heddle Award, *awarded annually to one doctoral candidate (from total cohort ~2000) at the University of Adelaide*

2014 Cooperative Research Centre for Landscape Evolution scholarship, *for outstanding research in physical geography*

2010 James Barrans prize for top graduating Earth Sciences major at the University of Adelaide

## Invited conference presentations

2027 **Falster, G.** & Hydroclimate2k Project Members: New insights into Common Era global water cycle variability from the Hydroclimate2k database. *Invited talk to be given in "*Multi-proxy perspectives of monsoon-to-polar climate connectivity*" session at the International Quaternary Association Congress, India*.

2025 **Falster, G.**, et al.: Paleoclimate-informed model simulations to test the link between tropical Indo-Pacific SST variability and mid-latitude megadroughts. *Invited talk in "*Climate of the Common Era*" session at the American Geophysical Union (AGU) Annual Meeting, USA*.Link to abstract.

2024 **Falster, G.**, et al.: New insights into the Pacific Walker Circulation from an 800-year-long water isotope-based reconstruction ensemble. *Invited talk in "*The atmospheric water cycle: processes, dynamics, isotopic tracers, and characteristics*" session at European Geosciences Union (EGU) General Assembly, Austria.*

2022 **Falster, G.**, et al.: No anthropogenic change in the frequency or severity of Australian multi-year droughts compared with the past millennium. *Invited talk at the Australian Research Council Centre of Excellence for Climate Extremes Annual Meeting, Australia*.

2020 **Falster, G.**, et al.: Fingerprinting the Pacific Walker Circulation using precipitation $\delta^{18}O$. *Invited talk in "*The atmospheric water cycle: processes, dynamics, isotopic tracers, and characteristics*" session at the AGU Fall Meeting, USA*.

### Invited book chapters

2026 **Falster, G.**, A. Atwood, B. Konecky, J. Tyler. *Climate of the Common Era*. Solicited chapter in the upcoming Elsevier Major Reference Work *Encyclopedia of Climate System Science*. Chapter submitted March 2026, book scheduled for publication October 2026.

### Selected invited seminars and keynote talks (international, funded)

2024 University of Cambridge (Department of Geography), UK
Tropical Hydroclimate Variability in the Quaternary (workshop), India (keynote speaker)
2023 Past Global Changes (PAGES) Topical Science Meeting, Germany (keynote speaker)
2021 University of Hawai'i (International Pacific Research Center), USA
Columbia University (Lamont-Doherty Earth Observatory), USA

## Leadership, service, and outreach

### Scientific leadership

2025– Hydroclimate2k project co-leader
*Facilitating compilation and interpretation of a comprehensive multi-archive multi-proxy global database of hydroclimate proxy records for the Common Era*
2022– Past Global Changes (PAGES) 2k Network Coordinator
*Planning and facilitating global collaborations to understand hydroclimate variability of the past 2000 years—informing anthropogenic changes to the climate system and hence likely future change.*
2020–2022 Scientific Program Committee for the PAGES Young Scientists Meeting

### Community leadership

2026– Selected to Australian-American Leadership Dialogue 'Young Leaders Dialogue'
*A diplomatic initiative bringing together emerging international leaders across government, enterprise, technology, media, scientific research, and education to review and refine the parameters of the Australian-American relationship.*

### Contributions to equity in STEM

2026– Mentor, L'Oréal-UNESCO For Women in Science PhD Mentoring Scheme
2025– Science and Technology Australia 'Superstar of STEM'. One of 60 female and non-binary STEM professionals selected to a) become high-profile role models inspiring future generations of female scientists and b) shift the public perception of STEM experts. *Link to profile*
2025– Regular collaborator with The Smith Family Foundation, giving career talks at public schools in low-income areas across Adelaide to de-mystify the pathway to becoming a scientist
2024– Participation in the 'Letters to a Pre-Scientist' program, where scientists are matched as pen pals with students from US schools in low-income areas, with the aim of fostering diversity in the next generation of climate scientists
2022–2023 ARC CLEX Culture & Diversity Committee
Mentor at Australian National University Women+ in STEM Leadership Conference
2022 Australian National University Research School of Earth Sciences Engagement Committee
2022 Co-convenor, Inclusivity Round Table at the PAGES Open Science Meeting
2020–2022 PAGES Open Science Meeting Diversity Committee

### Selected service to the academic community

2026– Steering committee for the Adelaide University Environment Institute Centre for Climate and Environmental Risk

2021– Session convenor at major international conferences
*Including at the AGU Fall Meeting, EGU General Assembly, INQUA Congress*

2019– Reviewer for *Nature, Nature Geoscience, Nature Communications, Journal of Hydrology, Geophysical Research Letters, Journal of Limnology, Boreas, Journal of Climate, Paleoceanography Paleoceanography and Paleoclimatology, Climate of the Past, Communications Earth & Environment.*

2023–2025 ARC Centre of Excellence for Climate Extremes (CLEX) Early Career Researcher Committee chair

2021–2024 PAGES Early Career Network Steering Committee

2016–2019 Australasian Quaternary Association (AQUA) Treasurer

2018 AQUA Biennial Conference local organising committee

### Selected outreach, policy, public science writing, and other contributions

2026 Invited speaker at South Australian Science Teachers Association Annual Meeting

2023–2024 *The Conversation* articles about atmospheric influences on extreme rainfall events and the possibility of Australian megadroughts

2023 European Geosciences Union invited blog post about the Pacific Walker Circulation. *Link to blog post. Shortlisted for EGU Best Blog Post of 2023*

2018 Australian National Science Week: exhibitor at 'Science in ACTion' school event

### Media presence

Radio, television, and online media presence regarding climate science topics including: global and regional weather and climate change; nature and causes of droughts; individual extreme weather events; long-term climate variability. See for example *Guardian article about my research into Australian droughts*

## Publications

**denotes paper led by one of my students*

### Journal articles in review

Lyu, Z, A. Cheung, Y. Bao, H. Zhang, Z. Tian, S. Zhao, Q. Dalaiden, P. Hopcroft, Q. Yin, A. Dallmeyer, J. Zhu, S. Tiwari, Z. Wu, Q. Wen, J. Smerdon, **G. Falster**, B. Konecky, F. Shi, M. Vuille: A mid-Holocene maximum in the Pacific Walker circulation. In review at *Nature*.

*Cho, P, **G. Falster**, D. Bolster, M. Müller: Walker Circulation interaction and their influence on tropical hydroclimate variability. In review at *Journal of Climate*.

Gillett, Z., S. Power, A. Taschetto, **G. Falster**, A. Ukkola: Global Projections of Multi-year Meteorological Drought in CMIP6 Simulations. In review at *Earth's Future*.

### Published peer-reviewed journal articles

**Falster, G.**, G. Abramowitz, S. Hobeichi, C. Hughes, P. Treble, N. Abram, M. Bird, A. Cauquoin, B. Dixon, R. Drysdale, C. Jin, N. Munksgaard, B. Proemse, J. Tyler, C. Tadros (2026). High resolution monthly precipitation isotope estimates across Australia from machine learning. *Hydrology and Earth System Science* **30**, 289–315. Link to publication.

Stevenson, S., C. Deser, S. Coats, **G. Falster**, B. Konecky, N. Maher, C. Pfleger (2026). Present-day tropical precipitation and cloud feedbacks determine future equatorial Pacific trends. *Science Advances* **12**(10), eaea8070. Link to publication.

Abram, N. J., N. Maher, S. Perkins-Kirkpatrick, **G. Falster**, T. Hughes, K. Meissnerm L. Slater, A. King, A. Pitman, G. Moon, W. Morgan (2025). Quantifying the regional to global climate impacts of individual fossil fuel projects to inform decision-making. *npj Climate Action* **4**, 92. Link to publication.

Vaiglova, P., H. Kierdorf, C. Witzel, **G. Falster**, R. Joannes-Boyau, Y. Wang, J. Wu, I. Williams, B. Knowles, Y. Wu, P. Bangsgaard, L. Yeomans, T. Richter, H. Darabi: Transport of animals underpinned ritual feasting at the onset of the Neolithic in southwestern Asia (2025). *Communications Earth & Environment* **6**, Article number 519. Link to publication.

*Cho, P., **G. Falster**, D. Bolster, M. Berke, M. Müller: Influence of the Indian Walker Circulation on $\delta^{18}O_P$ and Hydroclimate Variability in the Indian Ocean Basin (2025). *JGR Atmospheres* **138**(8), e2025JD043840. Link to publication.

Holgate, C., **G. Falster**, Z. Gillett, P. Goswami, M. Grant, S. Hobeichi, D. Hoffman, X. Jiang, C. Jin, X. Lu, M. Mu, J. Page, T. Parker, E. Vogel, N. Abram, J. Evans, A. Gallant, B. Henley, J. Kala, A. King, N. Maher, H. Nguyen, A. Pitman, S. Power, S. Rauniyar, A. Taschetto, A. Ukkola (2025). Physical mechanisms of meteorological drought development, intensification and termination: an Australian review. *Communications Earth and Environment* **6**, Article 220 (2025). Link to publication.

**Falster, G.,** S. Coats, N. Abram (2024). How Unusual Was Australia's 2017–2019 Tinderbox Drought? *Weather and Climate Extremes* **46**, 100734. Link to publication.

**Falster, G.**, N. Wright, N. Abram. A. Ukkola, B. Henley (2024). Potential for historically unprecedented Australian droughts from natural variability and climate change. *Hydrology and Earth System Science* **28**, 1383-1401. Link to publication.

Devanand, A., **G. Falster**, Z. Gillett, S. Hobeichi, C. Holgate, C. Jin, M. Mu, T. Parker, S. Rifai, K. Rome, M. Stojanovic, E. Vogel, N. Abram. G. Abramowitz, S. Coats, J. Evans, A. Gallant, A. Pitman, S. Power, S. Rauniyar, A. Taschetto, A. Ukkola (2024). Australia's Tinderbox Drought: an extreme natural event likely worsened by human-caused climate change. *Science Advances* **10**, eadj3460. Link to publication.

Konecky, B., N. McKay, **G. Falster**, S. Stevenson, M. Fischer, A. Atwood, D. Thompson, M. Jones, K. DeLong, J. Tyler, B. Martrat, E. Thomas, J. Conroy, S. Dee. L. Jonkers, O. Curakova (Sidorova), Z. Kern, T. Opel, H. Sayani, Iso2k Project Members (2023) Globally coherent water cycle response to temperature change during the past two millennia. *Nature Geoscience* **16**, 997-1004. Link to publication.

**Falster, G.**, B. Konecky, S. Coats, S. Stevenson. (2023). Forced changes in the Pacific Walker Circulation over the past millennium. *Nature* **662**, 93-100. Link to publication.

Churakova-Sidorova, O., V. Myglan, M. Fonti, O. Naumova, A. Kirdyanov, I. Kalugin, V. Babich, **G. Falster**, E. Vaganov, R. Siegwolf, M. Saurer (2023). Modern aridity in the Altai-Sayan mountain range derived from multiple millennial proxies. Scientific Reports **12**(1), 1-10. Link to publication.

**Falster, G.**, B. Konecky, M. Madhavan, S. Stevenson, S. Coats (2021). Imprint of the Pacific Walker Circulation in global precipitation $\delta^{18}O$. *Journal of Climate* **34**(21), 8579-8597. Link to publication.

Konecky, B., N. McKay, O. Churakova, L. Comas-Bru, E. Dassié, K. DeLong, **G. Falster**, M. Fischer, M. Jones, L. Jonkers, D. Kaufman, G. Leduc, S. Managave, B. Martrat, T. Opel, A. Orsi, J. Partin, H. Sayani, E. Thomas, D. Thompson, J. Tyler, N. Abram, A. Atwood, J. Conroy, Z. Kern, T. Porter, S. Stevenson, L. von Gunten, Iso2k Project Members (2020). The Iso2k Database: A global compilation of paleo-$\delta^{18}O$ and $\delta^{2}H$ records to aid understanding of Common Era climate. *Earth System Science Data* **12**, 2261-2288. Link to publication.

Khider, D., J. Emile-Geay**,** N. McKay, Y. Gil, D. Garijo, V. Ratnakar…[see paper for full authorship team]…**G. Falster** et al. (2019). PaCTS 1.0: a crowdsourced reporting standard for paleoclimate data. *Paleoceanography and Paleoclimatology* **34**(10), 1570-1596. Link to publication.

**Falster, G.**, S. Delean, J. Tyler (2018). Hydrogen peroxide treatment of natural lake sediment prior to carbon and oxygen stable isotope analysis of calcium carbonate. *Geochemistry, Geophysics, Geosystems* **19**(9), 3583-3595. Link to publication.

**Falster, G.,** J. Tyler, K. Grant, J. Tibby, C. Turney, G. Jacobsen, A. P. Kershaw: Millennial-scale variability in south-east Australian hydroclimate between 30,000 and 10,000 years ago (2018). *Quaternary Science Reviews* **192**, 106-122. Link to publication.

Collins, A., S. Patranabis-Deb, E. Alexander, C. Bertram, **G. Falster**, R. Gore, J. Mackintosh, P. Dhang, D. Saha, J. Payne, F. Jourdan, G. Backe, G. Halverson, B. Wade (2015). Detrital mineral age, radiogenic isotopic stratigraphy and tectonic significance of the Cuddapah Basin, India. *Gondwana Research* **28**(4), 1294-1309. Link to publication.

## Recent conference presentations (full list available on request)

2026 **Falster, G.**, G. Abramowitz, S. Hobeichi, C. Hughes, P. Treble, N. Abram, M. Bird, A. Cauquion, B. Dixon, R. Drysdale, C. Jin, N. Munksgaard, B. Proemse, J. Tyler, M. Werner, C. Tadros: Tropical Pacific imprint in water isotope variability across Australia. *Australian Meteorological and Oceanographic Society Conference, Australia* (oral).

**Falster, G.**, N. Abram, S. Wong, Z. Gillett: Palaeoclimate-informed model simulations to test the link between tropical Indo-Pacific SST variability and mid-latitude megadroughts. *Australian Meteorological and Oceanographic Society Conference, Australia* (oral).

2025 **Falster, G.**, M. Jones, N. McKay, A. Atwood, C. Hancock, B. Henley, N. Kaushal, M. Morales, A. Orsi, S. Shah, L. Smyth: Hydroclimate2k: A New PAGES Database of Hydroclimate Proxy Records for the Common Era. *AGU Annual Meeting*, *USA* (poster). Link to abstract.

**Falster, G.**, M. Jones, N. McKay, A. Atwood, C. Hancock, B. Henley, N. Kaushal, M. Morales, A. Orsi, S. Shah, L. Smyth: Hydroclimate2k: A New PAGES Database of Hydroclimate Proxy Records for the Common Era. *AGU Annual Meeting*, *USA* (poster). Link to abstract.

**Falster, G.**, G. Abramowitz, S. Hobeichi, C. Hughes, P. Treble, N. Abram, M. Bird, A. Cauquoin, B. Dixon, R. Drysdale, C. Jin, N. Munksgaard, B. Proemse, J. Tyler, C. Tadros: High resolution monthly precipitation stable isotope estimates across Australia from machine learning. *AGU Annual Meeting*, *USA* (poster). Link to abstract.

**Falster, G.**, N. Abram, S. Coats: The role of Indo-Pacific SST variability in global drought occurrence. *ACCESS-NRI meeting, Australia* (oral). Link to abstract and model set-up.

2024 **Falster, G.**, S. Coats: Assessing rainfall deficits during the 2017-2019 eastern Australian drought, in the context of previous observations. *Australian Meteorological and Oceanographic Society Conference, Australia* (oral).

2023 **Falster, G.**, B. Konecky, S. Stevenson, S. Coats: Anthropogenic impacts on the Pacific Walker Circulation: not quite what we expected. *AGU Fall Meeting, USA* (oral).

**Falster, G.**, H. Sayani, A. Orsi, H. McGregor, N. Kaushal, L. Jonkers, M. Jones, B. Henley, S. Eggleston, A. Atwood: Phase 4 of PAGES 2k: Hydroclimate of the Common Era. *EGU General Assembly, Austria* (oral; given in lieu by N. Kaushal).

## Selected non-traditional publications

2024 **Falster, G.** and PAGES Early-Career Network Steering Committee Members: What defines an "early-career researcher" in the paleosciences? *PAGES Magazine* **32**(1). Link to article

Co-author of annual Australian 'The State of Weather and Climate Extremes 2023' report *Links to report and my contribution*

2023 **Falster, G.**: 'Big Air Circle Ballad' *Science poem published in Consilience Journal* **15.**

2022 Evans, M., B. Valero Garcés, C. van Rensburg, Y. Ait Brahim, N. Schafstall, **G. Falster**, K. Meissner: Toward a more inclusive and diverse PAGES community. *PAGES Magazine* **30**(2). Link to article

2019 Amirebrahimi, S., K. Picard, N. Quadros, **G. Falster**: Multibeam Echo Sounder Data Acquisition in Australia and Beyond. Geoscience Australia Records 2019/08. Link to report

## Teaching and supervision

### Summary of undergraduate and postgraduate teaching

2025– At Adelaide University, I co-teach GEOLOGY3505 (*Earth Systems History*, third-year) and GEOLOGY2505 (*Geochemistry*, second-year).

2022–2024 At the Australian National University I co-taught EMSC1006 (*The Blue Planet,* first-year) and EMSC3019 (*Coral Reef Field* Studies, third-year). In both cases I taught in both the classroom and in the field. Over the three years, my Student Evaluation of Learning and Teaching scores for overall teaching quality averaged 4.7 out of 5.

2022–2024 At the Australian National University I guest lectured for ARCH8032 (*Introduction to Archaeological Science*) and EMSC4122 (*Analytical Methods*). In both postgraduate-level courses, I provided students with expert tuition in palaeoclimatology and water isotope systematics.

2023 ARC Centre of Excellent for Climate Extremes Winter School. I co-collaborated with climate science experts from four universities to each this intensive course for Australian climate science graduate students.

2019 At Washington University in St. Louis I guest lectured for EPSc486 (*Paleoclimatology*, fourth-year).

### PhD supervision

2026– S. Roth, The Australian National University (PhD, associate supervisor), *Understanding the climate reconstruction potential of wood anatomical and chemical characteristics of tree species across northern Queensland.*

2025– A. Matthias, Adelaide University (PhD, associate supervisor), *Understanding the long-term climate history of Kangaroo Island (Australia) with new speleothem records*

2023– O. Medd, The Australian National University (PhD, associate supervisor), *Developing and applying novel dendrochronological methods to understand snowgum dieback in the Australian alpine region*

2023– J. Harrison, The University of Melbourne (PhD, associate supervisor), *Developing and testing novel data assimilation algorithms for direct integration of palaeoclimate and climate model data*

### Undergraduate researcher supervision

Across 2021–2025 I supervised or co-supervised seven Honours and undergraduate earth science research projects, at Washington University in St. Louis and the Australian National University. I am currently the primary supervisor of an earth science Honours student at Adelaide University.